

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT	(a)	(i)		Tick boxes 1,4,5: Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto (1) Mars and the Earth have the same day length (1) The Earth has the greatest density (1) 4 boxes (2) 5 boxes (1) all boxes (0)			3	3		
FT	(b)	(i)		Longer wavelength						
		(ii)		Frequency increases						
		(iii)		Moves <u>away faster</u> Do not accept 'further away'						
1 HT	(c) FT (a) HT	(i)		Temperature = within range of -109 to -66(1) orbital time = within range 688 to 4330 (1)			2	2	2	
		(ii)		{Mass / diameter} smaller than the moon / smallest mass / smallest diameter accept 'smallest size' do not accept 'it is small/smaller'			1	1	1	
		(iii)		Scales: x-axis in intervals of 1/ 2 cm and y-axis in intervals of 10 / 2 cm (1) 5 plots correct $\pm$ less than one square tolerance (2) 4 plots correct $\pm$ less than one square tolerance (1) 3 or fewer plots correct $\pm$ less than one square tolerance (0) smooth curve (1) ignore extrapolation Do not accept thick/sketchy/wispy lines Do not award marks if incorrect variables are plotted or planets in incorrect order or bar chart If x-axis is labelled 'planets' do not award scale mark Moon plotted = neutral		4		4	4	
		(iv)		Orbital velocity decreases with distance / this graph curves downwards(1) (To be proportional) line must be a straight, upward and through origin (1)		2		2	2	

	(d) FT (b) HT	(i)	(Sunspots) are cooler (1) + Any 1 × (1) from: so do not emit as much light (1) OR (magnetic field) blocks hot gases reaching the surface (1)	1	1		2		
		(ii)	Electricity supply could be affected/power cut (1) TV / radio / mobile phone/satellite/communications affected (1)	2			2		
		(iii)	Number of sunspots (in each cycle) is different (1) and flares are linked with sunspots (1)		2		2		
	(e) FT (c) HT		Statements about one model implies the other model is different: Any 4 × (1) from: Aristotle: (converse for 2006 model) Earth at the centre of the Solar System / geocentric(1) Sun and planets orbit the Earth (1) Only one moon (1) Less planets (Absence of Uranus / Neptune / Plutoids) (1) Absence of asteroid belt (1)			4	4		
	(d) HT	(i)	Lines are red-shifted (1) All galaxies moving <u>away</u> (from the Earth)(1) The more distant the galaxy the faster it moves (1)						
		(ii)	{Same / similar} pattern of lines(1) So {same / similar} chemical composition(1) As each set of lines corresponds to an element (1)						
			Question 6/1 total	3	11	11	25	9	0